Statistics:

Accepted PhD average GRE: 165.14492753623188 Accepted Masters average GPA: 3.73491056910569
GRE vs GRE Verbal correlation: -0.07923377450388659

p-value: 0.10574205864954073

GPA vs GRE correlation: -0.003346400115644098
p-value: 0.9456163781687562

T-statistic: 0.12123930569647162
p-value: 0.9035145714287962

Analysis:

1. Average GRE score of an applicant who has been accepted into a PhD program

The average GRE score for applicants accepted into PhD programs was 165.14 (after filtering for valid GRE values greater than zero). This value falls within a realistic GRE range and reflects the subset of accepted PhD applicants rather than the full dataset, which contains many invalid or zero entries.

2. Average GPA of an applicant who has been accepted into a Master's program

The average GPA for applicants accepted into Master's programs was 3.73 (after filtering for valid GPAs between 0 and 4.0). This value aligns with expectations for competitive graduate admissions and is significantly higher than the overall dataset average due to the removal of invalid or zero GPA entries.

3. Shortcomings of inferring decision_date

Inferring decision_date from text introduces several limitations. First, notification formats are inconsistent, making parsing unreliable. Second, the year must be inferred from the term column, which can lead to incorrect assumptions if the term is missing or ambiguous. Third, many entries lack sufficient information to extract a valid date, resulting in missing values.

4. Correlation analysis

The correlation between GRE and GRE Verbal was -0.0792 ($p = 0.1057$), indicating a very weak, statistically insignificant relationship. The correlation between GPA and GRE was -0.00335 ($p = 0.9456$), indicating essentially no relationship. Overall, GPA and GRE-related measures do not appear to be meaningfully correlated in this dataset.

5. Group comparison (Accepted vs Rejected GPA)

A two-sample t-test comparing GPA between accepted and rejected applicants produced a test statistic of 0.1212 and a p-value of 0.9035. Since the p-value is much greater than 0.05, there is no statistically significant difference between the two groups, suggesting GPA alone does not distinguish acceptance outcomes in this dataset.

6. Chi-square test

The chi-square test comparing degree and country of origin produced a statistic of 93.26 with a p-value of 5.62×10^{-21}, indicating a strong statistically significant association. This suggests that the distribution of applicants across degree types differs meaningfully between American and international applicants.

7. Remaining data quality issues

Despite cleaning, the dataset still contains unrealistic values such as GPAs above 4.0 and GRE values outside valid ranges. Additionally, many entries contain zero values that likely represent missing data. Missing or incomplete date information also limits the reliability of time-based analyses.